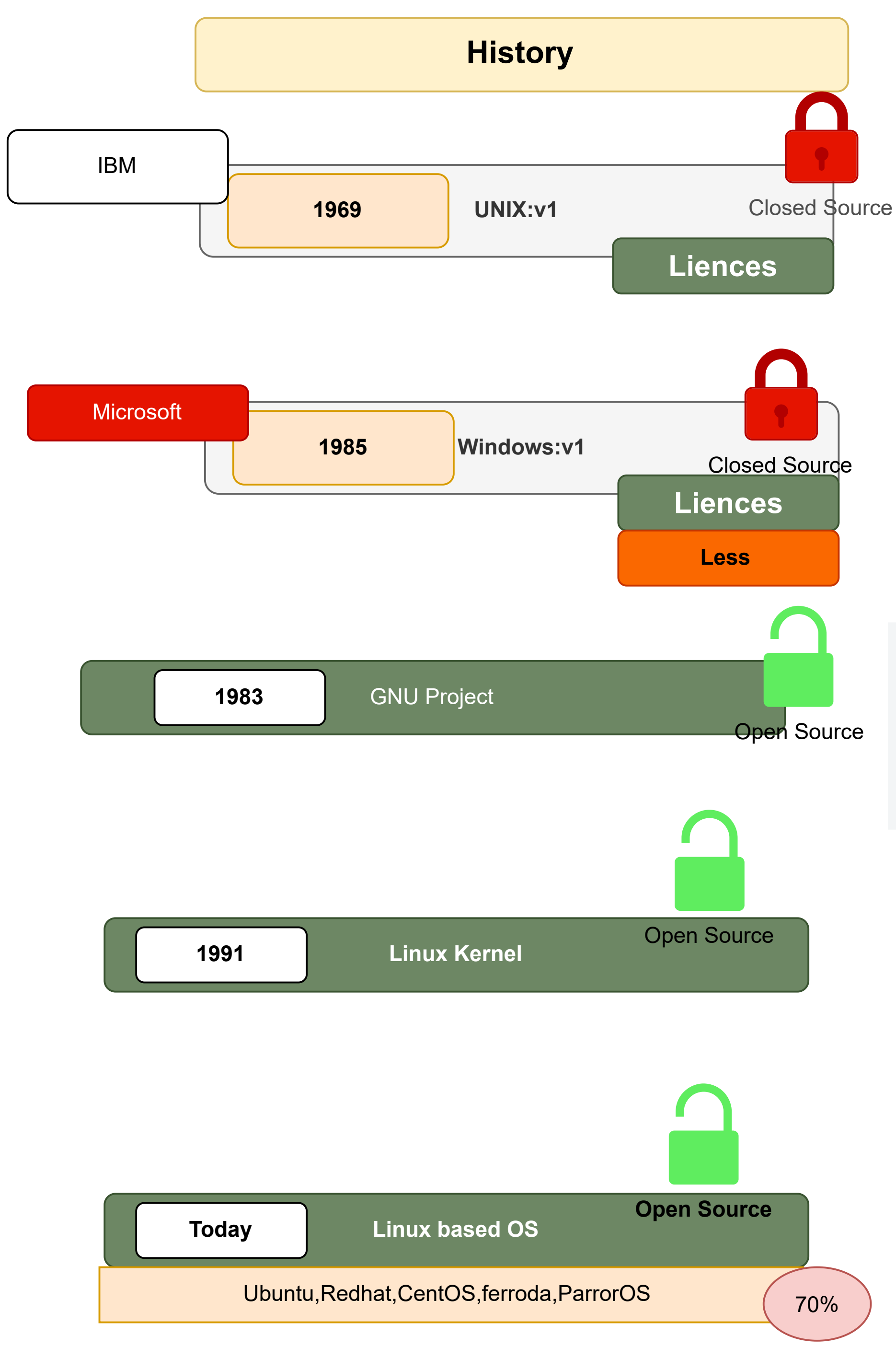
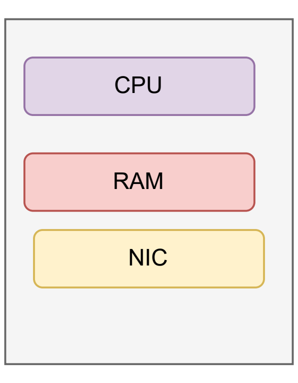
Linux Fundamentals – History, Kernel & Operating System

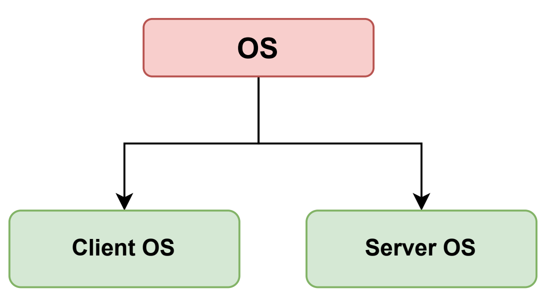
History



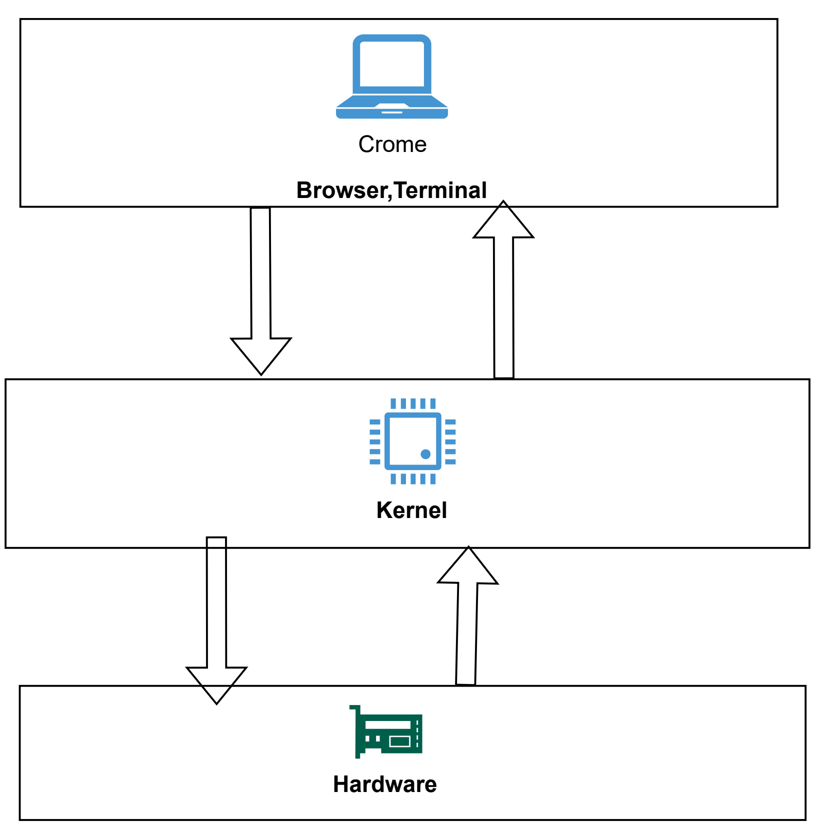


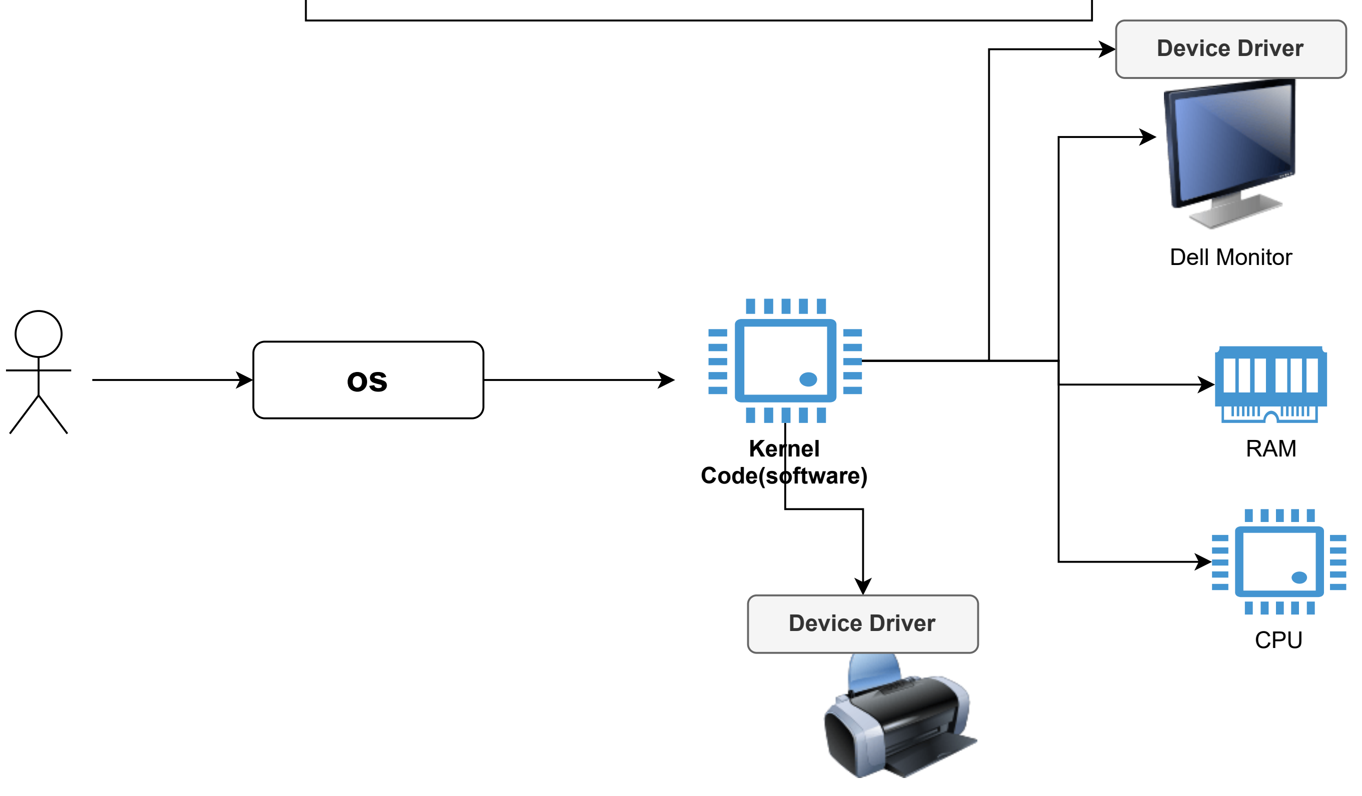
Richard Stallman

Linus torvalds

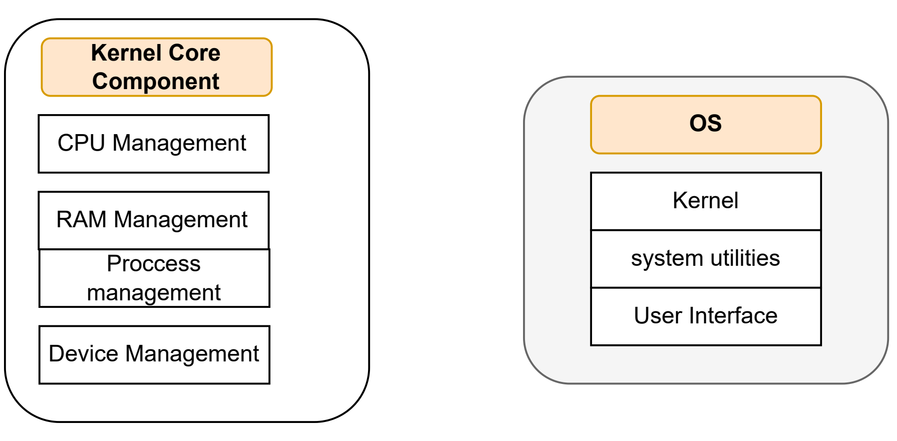


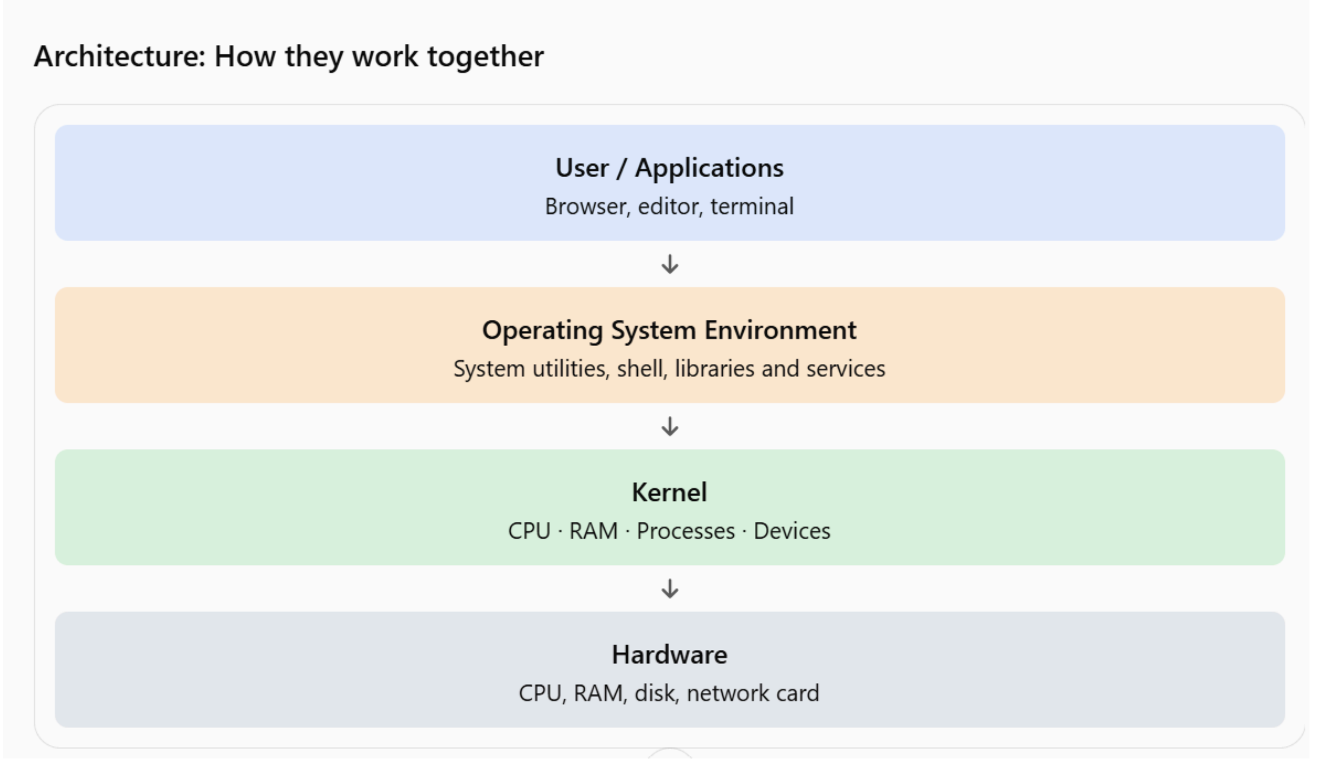
Linux Kernel





Kernel Vs os




Architecture: How they work together

User / Applications
Browser, editor, terminal

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Operating System Environment
System utilities, shell, libraries and services

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Kernel
CPU · RAM · Processes · Devices

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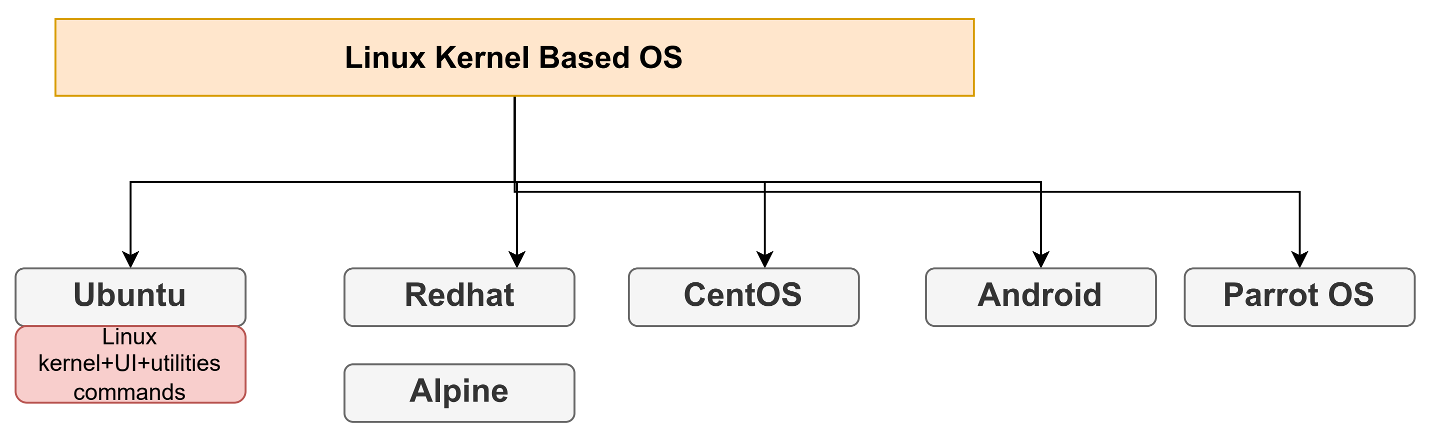
Hardware
CPU, RAM, disk, network card

Linux Kernel Based OS



1. Linux-Based OS List with Uses

Linux OS / Distribution	Kahan use hota hai?	Real-world example
Ubuntu Ubuntu Documentation	Beginners, developers, cloud, servers	Azure VM, web server, Docker host
Red Hat Enterprise Linux (RHEL) Red Hat	Enterprise production environments	Banking, corporate servers, enterprise applications
CentOS Stream	RHEL development ecosystem aur testing	Enterprise software testing, platform development
AlmaLinux	Enterprise Linux servers	Web hosting, production workloads
Rocky Linux	Enterprise Linux environments	Application servers, infrastructure labs
Debian	Stable servers aur general-purpose computing	Web servers, cloud infrastructure
Fedora	Developers aur newer Linux technologies	Development workstation, technology testing
Alpine Linux Alpine Linux	Lightweight containers	Docker images, microservices
Kali Linux Kali Linux Documentation	Cybersecurity aur penetration testing	Security labs, authorized vulnerability assessment
Parrot OS Parrot Linux	Security testing, digital forensics, privacy	Ethical hacking lab
Linux Mint	Desktop users aur beginners	Personal computer, office desktop
Arch Linux	Advanced users, customization	Custom Linux workstation
openSUSE / SUSE Linux Enterprise	Enterprise systems aur administration	Business servers, enterprise infrastructure
Amazon Linux	AWS cloud workloads	EC2 instances, cloud applications
Oracle Linux	Enterprise workloads aur Oracle ecosystem	Database servers, business applications
Raspberry Pi OS	Education, IoT, embedded projects	Raspberry Pi, hardware automation
Android Red Hat	Mobile devices	Smartphones, tablets, Android TV
ChromeOS	Chromebook devices	Education, browser-based computing

Linux Kernel Free & Secure